Geometric Limits on Semantic Expressibility: Quantum-Inspired Contexts and Polysemy in Large Language Models

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The geometric argument uses two counts. The *exact dimension* d is the number of mutually orthogonal axes in the usable representation space. The *quasi-dimensional count* N_{q} is the number of semantic directions represented with small, but nonzero, mutual overlap. Useful quasi-dimensionality lies between two failures. If $N_{\mathrm{q}} \le d$, exact orthogonal axes remain available and overcomplete encoding is unnecessary. If N_{q} is too large, the largest pairwise overlaps become too strong. For random-like directions and tolerated overlap ε, the resulting Goldilocks estimate is

$$d < N_{\mathrm{q}} \lesssim \exp(d\varepsilon^2/4).$$

Only a subset of size k is active at once; its readout interference scales as $\sqrt{k/d}$. Here ε and k are controls, not additional dimensions. On this geometric foundation we give a quantum-inspired but classically implementable model of polysemy: context observables share a token-associated spectral projector while acting differently, and generically noncommutatively, on sense subspaces in its orthogonal complement. Conditional projection onto a selected observable supplies a contextual embedding. The Goldilocks estimate and the intertwined-observable construction are separately testable and require no claim of physical quantum computation.

I. INTRODUCTION

This chapter examines contextual representation in large language models (LLMs) as part of the volume's broader inquiry into "quantum thinking." Only a minimal orientation to LLM operation is needed here. Input text is first divided into tokens, and each token is mapped to an initial vector. Transformer blocks then use self-attention and feed-forward transformations to turn those initially context-free vectors into sequence-dependent hidden states. An output layer converts the final state into token scores, and a softmax distribution supports probabilistic next-token selection. This compact chain—tokens, embeddings, contextual transformation, and finite readout—supplies the computational background for the argument below; the details of pre-training, fine-tuning, reinforcement learning, and backpropagation are not required.

Current Transformer-based LLMs are classical computational systems. They run on classical hardware and use real-valued matrix operations, normalization, nonlinearities, copying, and generally irreversible updates. Their reliance on vectors, inner products, high dimension, or probabilistic output does not by itself make them quantum or even quantum-like. The more limited question pursued here is where ordinary high-dimensional geometry already explains the coexistence and contextual selection of semantic features, and where a stronger comparison with quantum-like formalisms might begin. Such a comparison would have to do more than rename familiar neural operations: it would need to identify a role for contextual bases or subspaces, incompatible or noncommuting alternatives, interference, or measurement-like readout that yields explanatory or predictive value beyond a classical dynamic-embedding account.

Semantic concurrency makes this boundary concrete. A model must preserve many latent possibilities while making only a small, context-selected subset operationally accessible. Polysemy is the simplest illustration: the same token can participate in financial, river-edge, or seat/bench contexts of "bank," yet the surrounding sequence drives it toward different contextual representations. The verb in an aircraft "banks" supplies a fourth sense, mentioned below but not used in the three-arm schematic. This relation among possibility, context, and readout connects directly to the volume's recurring themes without implying that the underlying machine is physically quantum. It also motivates a separation between what a representation can geometrically accommodate and what a finite channel can reliably recover.

The claims below occupy different rungs of that evidential ladder. Concentration of measure and exponential packing are standard results of high-dimensional geometry. Their use to describe semantic capacity, together with the accumulated-interference estimate, is a formal model of classical representation subject to stated assumptions about the usable carrier and coactivation. The reference-subspace organization is a stronger empirical hypothesis about LLM internals and is tested against classical controls. The comparison with quantum contextuality remains a limited structural correspondence; no physical quantumness, Born-rule dynamics, or semantic Kochen–Specker theorem is inferred.

A finite-dimensional activation space faces a two-sided tension. The exact dimension d is the number of mutually orthogonal axes available in the usable carrier.

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The second count, N_q, is the number of semantic directions treated as quasi-dimensions. If $N_\mathrm{q} \le d$, exact orthogonality remains available. Once $N_\mathrm{q} > d$, the code is overcomplete and quasi-orthogonality becomes useful. Yet increasing N_q still further eventually produces too many large pairwise overlaps. This is the Goldilocks zone between an underloaded exact-axis regime and an overcrowded quasi-dimensional regime.

The two sides obey different scalings. On the capacity side, concentration of measure permits exponentially many candidate semantic directions with pairwise overlap at most ε,

$$N_\mathrm{q} \lesssim \exp(d\varepsilon^2/4). \tag{1}$$

On the readout side, the same small residual overlaps add in quadrature over an active set of size k, giving an interference scale

$$\sigma_\mathrm{int} \sim \sqrt{k/d}. \tag{2}$$

The first relation estimates how many quasi-dimensions fit into d exact dimensions at overlap tolerance ε. The second estimates the readout noise when k of them are active. Neither ε nor k is another dimension: ε is an overlap tolerance and k is an active count. Together the relations make d a *semantic concurrency bandwidth*.

Polysemy is the cleanest example. A single lexical item, such as “bank”, can access several mutually exclusive meanings, only one of which is selected in a given context. The same tension appears more generally in feature superposition, role binding, and compositional semantics [1]: many latent semantic degrees of freedom must coexist, while only a context-selected subset is read out. We use polysemy as a controlled entry point because it makes contextual selection explicit. In lexical semantics, polysemy is distinguished from homonymy by the relatedness of senses [2]. Modern Transformer-based LLMs implement contextual selection through dynamic contextual embeddings: a token representation is repeatedly transformed by self-attention and MLP layers as it absorbs information from its sentential environment [3].

The chapter develops two central ideas. First, concentration permits a large semantic dictionary, but both dictionary crowding and accumulated interference delimit a quantitative Goldilocks bracket. This geometric claim is architecture-independent once d, N_q, the overlap tolerance ε, and the active count k have been operationally defined. Second, for some canonically polysemous tokens, contexts may be represented by intertwined symmetric operators. These operators share a stable token-associated spectral projector, while sense information occupies low-dimensional subspaces in its orthogonal complement. Selecting or mixing the corresponding conditional projections implements a contextual embedding. This operator construction is not implied by concentration geometry; it is a stronger empirical model, and Sec. V D states how it can fail.

A serious caveat is anisotropy. The nominal width of an LLM layer may exceed the dimension of the carrier actually occupied by its states. To avoid a third dimensional symbol, d throughout this chapter means the *usable exact dimension*: first restrict or whiten the representation to the carrier being tested, then let d be the size of an orthonormal basis of that carrier. If this usable d is small, the quasi-dimensional capacity weakens. The method used to identify the carrier must therefore be declared empirically.

The reference-subspace and operator hypotheses have a narrower domain. They are meant for canonically polysemous tokens with established sense inventories, such as WordNet-, OntoNotes-, SemCor-, or WiC-style sense distinctions [4] with enough contextual instances for testing. Failure on a representative sample, against the controls in Sec. V D, would count against a privileged token reference direction or an intertwined-observable implementation even if the broader Goldilocks capacity/accessibility picture survives.

The proposal is adjacent to the superposition hypothesis in mechanistic interpretability [5], according to which neural networks represent more features than they have neurons by packing features into nearly orthogonal directions. The present chapter separates the geometric background from a possible semantic organization within it. The general superposition picture concerns an overcomplete dictionary of feature directions and sparse active sets. The present model adds the more specific possibility that contextual readout is organized around stable token-associated directions and context-dependent, potentially noncommuting projectors.

Section II develops the capacity framework: quasi-orthogonality, Lévy’s lemma, exponential packing, and the kinetic/epistemic distinction, culminating in the Goldilocks bracket of Sec. II E. Section III introduces the reference-subspace hypothesis, its realization by intertwined context observables, and its relation to dynamic contextual embeddings. Section IV connects the construction to attention as soft context routing rather than literal basis reconstruction. Section V discusses superposition, sparse autoencoders, limitations, empirical tests, and the limited quantum analogy.

The two dimensional counts. Throughout, $\mathbb{R}^d$ is the usable Euclidean carrier. Its exact dimension d is the size of any orthonormal basis. The symbol N_q denotes the number of approximately orthogonal semantic directions represented in that carrier; these are the paper’s *quasi-dimensions*. Thus N_q is a count of directions, not the dimension of a second vector space. Other symbols are controls or local ranks: ε is an overlap tolerance, k is the number simultaneously active, and the later p and q_C count basis vectors inside particular reference and sense subspaces.

The carrier has inner product $u^\mathsf{T} v$ and norm $\|v\| = \sqrt{v^\mathsf{T} v}$; $S^{d-1} = \{v \in \mathbb{R}^d : \|v\| = 1\}$ is the unit sphere; I is the identity matrix; $\|\cdot\|_\mathrm{op}$ is the operator norm; $A \succeq 0$ means positive semidefinite; and $\mathbb{P}$ and $\mathbb{E}$ denote probability and expectation under the uniform measure on the relevant sphere unless stated otherwise. Universal positive constants are denoted c, c', possibly changing from line to line.

II. CONCENTRATION GEOMETRY AND CAPACITY

This section is architecture-independent. It gives the geometric capacity constraint on which the later reference-subspace hypothesis builds.

A. Quasi-orthogonality

Let $w_1, \ldots, w_{N_\mathrm{q}} \in \mathbb{R}^d$ be unit semantic directions. Strict orthogonality, $w_i^\mathsf{T} w_j = 0$ for $i \neq j$, allows at most $N_\mathrm{q} = d$ such vectors. Semantic encoding does not require exact orthogonality. It requires approximate distinguishability. We therefore call a family ε-quasi-orthogonal if

$$|w_i^\mathsf{T} w_j| \leq \varepsilon, \qquad i \neq j. \tag{3}$$

We use "quasi-orthogonal" rather than "pseudo-orthogonal": the latter is normally reserved for geometry with an indefinite inner product, which is not intended here.

This is the relevant notion for high-dimensional feature storage: small overlap is tolerable, provided readout can absorb the resulting interference.

B. Lévy's lemma and linear overlaps

High-dimensional spheres make quasi-orthogonality generic.

The mechanism is already visible before invoking the formal lemma. Let

$$x = (x_1, \ldots, x_d) \in \mathbb{R}^d$$

be a unit vector chosen without a preferred direction. Then

$$x_1^2 + \cdots + x_d^2 = 1.$$

Rotational invariance suggests that the norm is spread over the available coordinates, so a typical coordinate weight is of order $1/d$. Now fix a unit direction u. By rotational invariance one may choose coordinates so that $u = e_1$. The squared overlap with the random direction is then

$$|u^\mathsf{T} x|^2 = x_1^2.$$

Thus the overlap with any fixed direction is typically of order $d^{-1/2}$ in amplitude, or $1/d$ in squared amplitude. A macroscopic overlap $|u^\mathsf{T} x| > \varepsilon$ therefore requires one coordinate to carry an anomalously large fraction of the total norm. In high dimension such an event is exponentially unlikely.

The same intuition appears particularly transparently in a d-dimensional complex Hilbert-space analogue. If $\psi \in \mathbb{C}^d$ is Haar-random and ϕ is fixed, unitary invariance permits the choice $\phi = e_1$, so

$$|\langle \phi, \psi \rangle|^2 = |\psi_1|^2.$$

For a Haar-random complex unit vector the exact tail is

$$\mathbb{P}\{|\langle \phi, \psi \rangle|^2 \geq \eta\} = (1-\eta)^{d-1} \leq \exp[-(d-1)\eta].$$

For real embedding vectors the analogous spherical concentration estimate has the operative scaling

$$\mathbb{P}\{|u^\mathsf{T} x| > \varepsilon\} \lesssim \exp(-cd\varepsilon^2)$$

for a universal constant $c > 0$. Consequently, if N_q random directions are sampled, a union-bound estimate gives a probability at most of order $N_\mathrm{q}^2 \exp(-cd\varepsilon^2)$ that some pair has overlap larger than ε. Hence N_q may grow exponentially in $d\varepsilon^2$ while all pairwise overlaps remain small with high probability. This is the heuristic core of the exponential quasi-orthogonal capacity derived below.

Lévy's lemma makes this intuition precise. If $f : S^{d-1} \to \mathbb{R}$ is L-Lipschitz, then

$$\mathbb{P}\big(|f(v) - \mathbb{E}f| \geq \delta\big) \leq 2\exp\left(-\frac{c\,d\,\delta^2}{L^2}\right), \tag{4}$$

for a universal $c > 0$ [6, 7]. Apply this to the linear functional $f_u(v) = u^\mathsf{T} v$. It is 1-Lipschitz and has mean zero by symmetry. Hence

$$\mathbb{P}\big(|u^\mathsf{T} v| \geq \varepsilon\big) \leq 2\exp(-c\,d\,\varepsilon^2). \tag{5}$$

For normalized surface measure one obtains the explicit form $2\exp(-(d-1)\varepsilon^2/2)$ [6]. Thus two random unit vectors are nearly orthogonal with high probability, and typical overlap amplitudes scale as $d^{-1/2}$.

In trained models, d means the usable carrier defined in the Introduction, not automatically the nominal layer width. Anisotropy or a narrow activation cone requires restriction or whitening before the orthonormal-basis size d is assigned.

Relation to Johnson–Lindenstrauss. The Johnson–Lindenstrauss lemma preserves pairwise distances among n points after projection to dimension $r = O(\varepsilon^{-2}\log n)$ [8]. Here r is only the temporary target rank in that theorem; it is unrelated to the active-set count k used below. The lemma is driven by the same concentration phenomenon as Eq. (5); indeed JL can be derived from sphere concentration applied to projection norms [9]. Here Lévy's lemma is the more direct tool: the issue is not projecting a finite dataset, but the typical overlap between semantic directions in the representation space.

C. Exponential packing

Sample N_q unit vectors independently and uniformly from S^{d-1}. By Eq. (5) and a union bound over pairs,

$$\mathbb{P}\big(\exists\, i \neq j :\ |w_i^\mathsf{T} w_j| \geq \varepsilon\big) < N_\mathrm{q}^2 \exp(-c\, d\, \varepsilon^2). \tag{6}$$

This probability is below one, and hence an ε-quasi-orthogonal family exists, provided

$$N_\mathrm{q} < \exp\Big(\frac{c}{2}\, d\, \varepsilon^2\Big). \tag{7}$$

This is the kinetic capacity estimate: for large usable exact dimension d, many candidate semantic directions can coexist with small pairwise overlap.

The estimate controls pairwise overlap only. It does not guarantee that a large collection of feature directions is well-conditioned. By Gershgorin, $|w_i^\mathsf{T} w_j| \leq \varepsilon$ gives only $\|G - I\|_\mathrm{op} \leq (n-1)\varepsilon$ for an n-element Gram matrix, which is useless when n is large. Joint readout therefore needs a separate conditioning assumption, introduced in Sec. III C.

D. Kinetic capacity versus epistemic accessibility

The packing bound says what can coexist. It does not say what can be read. We call the coexistence supply *kinetic semantic capacity*. We call the recoverable subset *epistemic semantic accessibility*. The distinction is the central point of the chapter.

Let N_q candidate features be represented by unit directions $w_1, \ldots, w_{N_\mathrm{q}} \in \mathbb{R}^d$. In a given context, suppose an active set S of size k contributes amplitudes x_j, producing

$$h = \sum_{j \in S} x_j w_j. \tag{8}$$

A linear readout of feature i gives

$$w_i^\mathsf{T} h = x_i + \sum_{\substack{j \in S \\ j \neq i}} x_j\, w_i^\mathsf{T} w_j. \tag{9}$$

The first term is signal; the second is interference. For random incoherent directions,

$$\mathbb{E}[w_i^\mathsf{T} w_j] = 0, \qquad \mathrm{Var}(w_i^\mathsf{T} w_j) \simeq \frac{1}{d}. \tag{10}$$

Thus, conditional on the active amplitudes,

$$\mathrm{Var}\left(\sum_{\substack{j \in S \\ j \neq i}} x_j\, w_i^\mathsf{T} w_j\right) \simeq \frac{1}{d} \sum_{\substack{j \in S \\ j \neq i}} x_j^2. \tag{11}$$

For amplitudes of order unity,

$$\sigma_\mathrm{int} \sim \sqrt{\frac{k}{d}}. \tag{12}$$

This is the readout bottleneck. Concentration lets many directions coexist because their overlaps are small. But readout sees the *sum* of many small overlaps. Reliable recovery requires

$$\sqrt{\frac{k}{d}} \ll \Delta, \tag{13}$$

where Δ is the relevant decision margin. Equivalently,

$$\text{kinetic: } N_\mathrm{q} \lesssim \exp(c\, d\, \varepsilon^2), \qquad \text{epistemic: } k \lesssim d\, \Delta^2. \tag{14}$$

The first is storage-like coexistence. The second is simultaneous accessibility. A model may host many more features than it can jointly read.

E. The Goldilocks zone in plain terms

Only two global geometric counts are needed. The exact dimension d is the number of vectors in an orthonormal basis of the chosen usable carrier. The quasi-dimensional count N_q is the number of semantic directions represented in that carrier. Strictly speaking, N_q is a dictionary size rather than the dimension of a second vector space. We nevertheless call its members "quasi-dimensions" when their mutual overlaps are small enough for the intended readout. Any strong anisotropy or unused coordinates must be handled when the usable carrier is chosen, before d is assigned; it does not introduce a third dimension symbol.

The three regimes can then be stated directly. If $N_\mathrm{q} \leq d$, an exactly orthogonal assignment is geometrically possible, although learning need not select one and no semantic basis is privileged. If $N_\mathrm{q} > d$, exact mutual orthogonality is impossible, but many approximately orthogonal directions may still coexist. Eventually the dictionary becomes so crowded that some directions overlap too strongly to remain reliably distinguishable.

A simple random-code estimate locates this transition. For N_q random-like unit directions in $\mathbb{R}^d$, the largest pairwise overlap is typically of order

$$\mu_\mathrm{max} := \max_{i \neq j} |w_i^\mathsf{T} w_j| \approx 2\sqrt{\frac{\log N_\mathrm{q}}{d}}. \tag{15}$$

The factor 2 reflects the fact that a dictionary contains of order N_q^2 pairs. If ε is the largest acceptable overlap, solving $\mu_\mathrm{max} \lesssim \varepsilon$ gives the transparent Goldilocks estimate

$$\boxed{d < N_\mathrm{q} \ \lesssim\ \exp\left(\frac{d\varepsilon^2}{4}\right)}. \tag{16}$$

The left boundary marks the onset of overcompleteness. The right boundary marks crowding under an isotropic random null model. There is no unique numerical upper boundary unless an overlap tolerance is chosen: ε is a distinguishability criterion, not another dimension.

Equivalently, supporting a prescribed quasi-dimensional count requires roughly

$$d \gtrsim \frac{4 \log N_{\mathrm{q}}}{\varepsilon^2}. \tag{17}$$

For example, let $d = 4096$. A single random pair then overlaps by about $1/\sqrt{d} = 1.6\%$ on its natural scale. But $N_{\mathrm{q}} = 10^4$ directions make almost fifty million pairs, so Eq. (15) estimates the largest overlap as about 9.5%. With a 10% tolerance, Eq. (16) gives the illustrative band

$$4096 < N_{\mathrm{q}} \lesssim 2.8 \times 10^4. \tag{18}$$

This number is not an LLM constant. It changes exponentially with the chosen tolerance, and learned directions are neither independent nor isotropic.

Simultaneous accessibility is a separate issue. If k directions are active at once, their residual interference has the scale $\sigma_{\mathrm{int}} \sim \sqrt{k/d}$ from Eq. (12). Here k is an active-set count, not a dimension. For $d = 4096$ and $k = 64$, the null-model interference scale is 0.125. Thus a dictionary can lie inside the geometric Goldilocks zone yet still be difficult to read when too many of its directions are activated together.

Figure 1 illustrates these two statements. It is not a model of a trained language model, but a null model for concentration geometry. Departures in trained representations would be informative about anisotropy, learned feature alignment, nonlinear readout, or coactivation-dependent orthogonalization.

This also clarifies the relation to compressed sensing. A dictionary of N_{q} feature directions with only k active at once can sometimes be read from far fewer than N_{q} coordinates. Under suitable incoherence or restricted-isometry assumptions, the familiar scaling is $d \gtrsim Ck \log(N_{\mathrm{q}}/k)$ [10–12]. The analogy is diagnostic rather than literal: hidden states are anisotropic, feature activation is structured, and transformer readout is nonlinear.

III. TOKEN REFERENCE SUBSPACES AND INTERTWINED CONTEXT OBSERVABLES

The previous section established an architecture-independent capacity constraint. We now ask a more speculative question: does a trained model organize the contextual states of some polysemous tokens relative to a stable token-associated reference subspace? This hypothesis is not implied by concentration geometry, and it is separately falsifiable.

The notation distinguishes mathematical types and roles. For a token t and a sense context C:

- $r_t \in \mathbb{R}^d$ is a unit candidate token reference vector;
- $\mathcal{R}_t \subset \mathbb{R}^d$ is the token reference subspace;
- P_t is the orthogonal projector onto $\mathcal{R}_t$;
- $S_{t,C} \subset \mathcal{R}_t^{\perp}$ is a low-dimensional sense subspace;
- $u_{t,C,j} \in S_{t,C}$ are vectors spanning that sense subspace;
- $h_\ell(t \mid x) \in \mathbb{R}^d$ is an observed contextual hidden state.

Thus r_t, $u_{t,C,j}$, and $h_\ell(t \mid x)$ are vectors, whereas $\mathcal{R}_t$ and $S_{t,C}$ are subspaces. The w_i used in Sec. II have a separate role: they are generic feature directions in the capacity calculation and are not reused as sense labels.

A. The reference-subspace hypothesis

In the simplest version, let r_t be a candidate unit direction and set

$$\mathcal{R}_t = \mathrm{span}\{r_t\}. \tag{19}$$

Empirically, r_t could be the normalized static input embedding, a normalized mean contextual embedding, or a learned token-specific probe direction. It is simply a hypothesized context-stable reference for token identity. It is not a mechanical or dynamical object. Any state can be decomposed relative to any direction, so the decomposition alone has no empirical content. The claim is that a token-associated $\mathcal{R}_t$ exposes sense structure better than matched control directions or subspaces.

For each context or sense label C, define a low-dimensional subspace

$$S_{t,C} = \mathrm{span}\{u_{t,C,1}, \ldots, u_{t,C,q_C}\} \subset \mathcal{R}_t^{\perp}, \qquad q_C \ll d. \tag{20}$$

For the intended three-arm example, finance, river-edge, and seat/bench associations of "bank" correspond to different subspaces $S_{\mathrm{bank},C}$ inside the common orthogonal carrier $\mathcal{R}_{\mathrm{bank}}^{\perp}$. The verbal use in which an aircraft banks or tilts supplies a fourth sense and would add a fourth context subspace; it is mentioned here but omitted from the drawing for clarity. A word sense is not a single vector: topic, syntax, selectional preferences, and world knowledge may require several spanning directions $u_{t,C,j}$.

The reference object may itself require more than one direction. In that case define

$$\mathcal{R}_t = \mathrm{span}\{r_{t,1}, \ldots, r_{t,p}\}, \qquad p \geq 1, \tag{21}$$

and require

$$S_{t,C} \subset \mathcal{R}_t^{\perp}. \tag{22}$$

Equation (19) is the special case $p = 1$. The bare subspace construction specifies only a pattern of sharing. In quantum logic, different contexts can share one or more observables or spectral projections; finite-dimensional examples with two shared observables are known [13]. In the semantic model, $\mathcal{R}_t$ is a common token-associated reference subspace, while the sense-specific subspaces lie

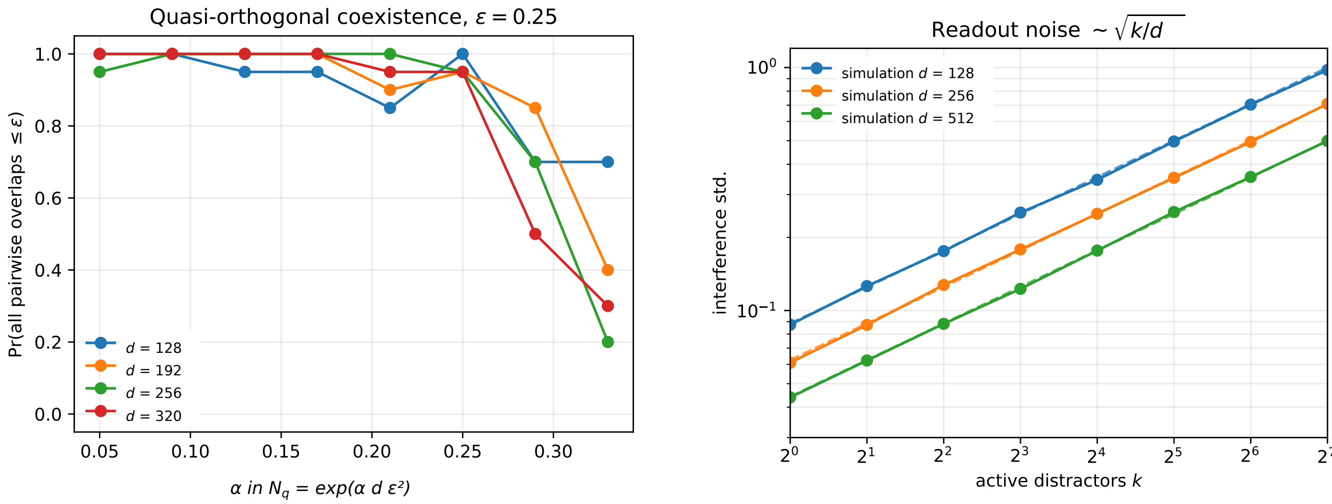

FIG. 1. Synthetic geometry checks for the capacity/accessibility distinction. Left: random unit directions in exact usable dimension d are sampled with $N_{\mathrm{q}} = \exp(\alpha d \varepsilon^2)$ and tested for ε-quasi-orthogonality. For sufficiently small α, large families coexist with high probability, illustrating the exponential packing side of Eq. (14). Right: for a fixed target direction, k random distractor directions are simultaneously active with random signs. The standard deviation of the readout interference follows the predicted scale $\sigma_{\mathrm{int}} \sim \sqrt{k/d}$, illustrating the accessibility side of Eq. (14). Together the panels visualize coexistence and simultaneous accessibility. Both use random isotropic vectors and should be read as a geometric null model rather than as measurements on a trained language model.

in its orthogonal complement. These incidence relations alone imply neither noncommutativity nor quantum probability. The next subsection makes the stronger, optional operator structure explicit.

The low-rank assumption is empirical. It predicts that dominant within-sense variation is low-dimensional after projection onto $\mathcal{R}_t^{\perp}$. This is compatible with evidence that contextualized embeddings can separate word senses, but it is stronger: it asks whether such separation is privileged relative to a token-associated reference subspace [14, 15].

B. Intertwined context observables and conditional embeddings

The shared-subspace idea can be implemented as a family of intertwined context observables. Let $Q_t \in \mathbb{R}^{d\times p}$ have orthonormal columns spanning $\mathcal{R}_t$, so that $P_t = Q_t Q_t^{\mathsf{T}}$. For each context C, let $B_{t,C} \in \mathbb{R}^{d\times q_C}$ have orthonormal columns spanning $S_{t,C}$. Here Q_t and $B_{t,C}$ are basis matrices, not semantic vectors.

The projector onto the token-plus-sense subspace is

$$\Pi_{t,C} = P_t + B_{t,C} B_{t,C}^{\mathsf{T}}. \tag{23}$$

All $\Pi_{t,C}$ contain the same spectral projector P_t and therefore intertwine on $\mathcal{R}_t$. Their sense parts differ. In particular,

$$[\Pi_{t,C}, \Pi_{t,C'}] = [B_{t,C} B_{t,C}^{\mathsf{T}}, B_{t,C'} B_{t,C'}^{\mathsf{T}}] \tag{24}$$

need not vanish. Nonzero commutators represent incompatible conditional decompositions: applying the two context projections in opposite orders can give different intermediate states.

A symmetric context observable with this eigenspace structure is, for example,

$$A_{t,C} = Q_t \Lambda_t Q_t^{\mathsf{T}} + B_{t,C} \Lambda_{t,C} B_{t,C}^{\mathsf{T}}, \tag{25}$$

where the diagonal matrix Λ_t is shared across contexts and $\Lambda_{t,C}$ is context-specific; their eigenvalue sets may be chosen disjoint so that P_t is identifiable as a shared spectral sector. The operator acts as zero on the unused orthogonal complement; an orthonormal completion may assign additional eigenvalues without changing the shared structure. With a nondegenerate completed spectrum, $A_{t,C}$ is the real finite-dimensional analogue of a maximal observable. Generic choices for different C need not commute, although every $A_{t,C}$ retains the common spectral sector P_t.

The corresponding context-conditioned embedding is concrete:

$$h_{t,C}(x) = \Pi_{t,C} h(x) = P_t h(x) + B_{t,C} B_{t,C}^{\mathsf{T}} (I - P_t) h(x). \tag{26}$$

One may select a single C, or take a soft mixture of these embeddings as in Sec. IV. Equations (23)– (26) provide the proposed implementation of intertwining observables for contextual embedding. They use ordinary real matrices and can be trained or probed in a classical network. Thus noncommutativity here is a testable representational property, not evidence of physical quantum dynamics or the Born rule.

C. Projection and gated subspace readout

Let $h(x) \in \mathbb{R}^d$ be a contextual state and retain the orthogonal projector P_t defined above. The two components are

$$\begin{aligned} h(x) &= h_{\mathrm{ref}}(x) + h_\perp(x), \\ h_{\mathrm{ref}}(x) &= P_t h(x), \\ h_\perp(x) &= (I - P_t)h(x) \in \mathcal{R}_t^\perp. \end{aligned} \tag{27}$$

For the one-dimensional case, $P_t = r_t r_t^\mathsf{T}$ and the scalar $a_t(x) = r_t^\mathsf{T} h(x)$ measures the component along the reference direction. The hypothesis is that most sense-disambiguating variation is present in $h_\perp(x)$, not that the projection formula itself is special.

Readout inside $S_{t,C}$ also requires the spanning vectors to be well-conditioned. Let

$$U_{t,C} = [\, u_{t,C,1} \ \cdots \ u_{t,C,q_C} \,], \qquad G_{t,C} = U_{t,C}^\mathsf{T} U_{t,C}. \tag{28}$$

The columns of $U_{t,C}$ span $S_{t,C}$; $U_{t,C}$ is a matrix, not an additional semantic vector. Unlike the orthonormal $B_{t,C}$ used in the spectral construction, $U_{t,C}$ may be a learned nonorthogonal spanning matrix. We call it ρ-well-conditioned if

$$\|G_{t,C} - I\|_{\mathrm{op}} \le \rho < 1. \tag{29}$$

Then

$$\widetilde{U}_{t,C} = U_{t,C} G_{t,C}^{-1} \tag{30}$$

is a dual spanning matrix and satisfies $\|\widetilde{U}_{t,C}\|_{\mathrm{op}} \le (1-\rho)^{-1/2}$. Near-degenerate spanning sets amplify readout noise, so the model requires ρ comfortably below one. The corresponding coordinates are

$$\alpha_{t,C}(x) = \widetilde{U}_{t,C}^\mathsf{T} h_\perp(x) \in \mathbb{R}^{q_C}. \tag{31}$$

A simple sense score is

$$s_{t,C}(x) = g_t(a_t(x))\, f_{t,C}(h_\perp(x)), \tag{32}$$

where $g_t \ge 0$ gates engagement with the token reference direction and $f_{t,C} \ge 0$ measures fit to the sense subspace. One possible choice is $f_{t,C} = \|\alpha_{t,C}(x)\|^2$; a learned positive quadratic form is another. The selected sense label is

$$C^*(x) = \arg\max_C s_{t,C}(x). \tag{33}$$

The empirical content is the proposed separation of roles: the reference component tracks token identity or engagement, while the orthogonal component contains sense-disambiguating information. If no token-associated $\mathcal{R}_t$ exposes sense structure better than matched random, frequency-controlled, or layer-matched subspaces, the hypothesis fails.

Figure 2 illustrates these objects as an abstract hypergraph incidence structure.

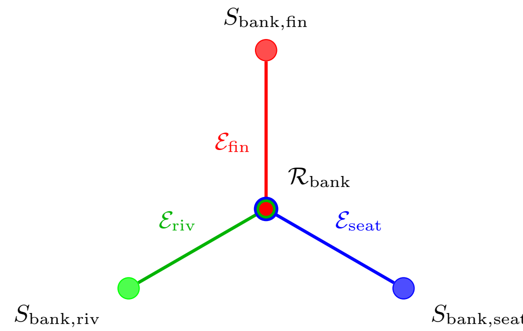

FIG. 2. Hypergraph schematic of context organization for the polysemous token "bank." The original star geometry is retained, but its elements are now typed as abstract hypergraph objects rather than spatial vectors. The central vertex denotes the shared token reference subspace $\mathcal{R}_{\mathrm{bank}}$; the colored outer vertices denote the complete low-dimensional sense subspaces $S_{\mathrm{bank},C} \subset \mathcal{R}_{\mathrm{bank}}^\perp$, not individual directions. Each colored spoke is the context hyperedge $\mathcal{E}_C = \{\mathcal{R}_{\mathrm{bank}}, S_{\mathrm{bank},C}\}$ for $C \in \{\mathrm{fin}, \mathrm{riv}, \mathrm{seat}\}$. Each hyperedge is the combinatorial shadow of a context projector $\Pi_{\mathrm{bank},C}$ or observable $A_{\mathrm{bank},C}$. Because each displayed hyperedge has two abstract vertices, it is drawn as an ordinary line; a context with additional components would be a higher-order hyperedge incident on all of them. The star is an incidence diagram, not a coordinate plot, and its spokes do not denote vector directions, maps, or causal flow.

D. Packing sense subspaces in a shared orthogonal carrier

The vector packing bound extends to low-dimensional subspaces. Fix a reference subspace $\mathcal{R}_t$ of rank p. Its orthogonal complement has exact dimension $d - p$. To state the comparison simply, suppose the candidate sense subspaces have a common local rank $q \ll d - p$. For any fixed threshold $\mu > 2\sqrt{q/(d-p)}$, there exist many such subspaces $S_1, \ldots, S_M \subset \mathcal{R}_t^\perp$ satisfying

$$\|U_a^\mathsf{T} U_b\|_{\mathrm{op}} \le \mu, \qquad a \ne b, \tag{34}$$

where $U_a, U_b \in \mathbb{R}^{d \times q}$ are orthonormal basis matrices whose columns lie in $\mathcal{R}_t^\perp$, and

$$M \ge \exp\big(c_{\mu,q}(d-p)\big) \tag{35}$$

for some positive $c_{\mu,q}$.

This is a standard Grassmannian packing consequence of concentration on Stiefel and Grassmann manifolds [16, 17]. For two independent Haar-random q-planes, the singular values of $U^\mathsf{T} V$ are the cosines of their principal angles. In the regime $q \ll d-p$, the largest principal correlation is of order $\sqrt{q/(d-p)}$ [18–20]. A union bound over sampled subspaces then gives an exponential packing.

Here p and q are local subspace ranks, and M counts candidate sense subspaces; none is a new global capacity dimension. For trained representations, the comparison assumes that the usable d-dimensional carrier has already been selected or whitened. Otherwise the Haar-random subspace model is not a good description of activation geometry.

E. Relation to dynamic contextual embeddings

Transformers already handle polysemy dynamically. The hidden state $h_\ell(t \mid x)$ of token t at layer ℓ depends on context x. The reference-subspace and intertwined-observable hypotheses are not alternative hardware architectures; they are proposed organizations and readouts of these observed trajectories.

Given a candidate reference subspace $\mathcal{R}_t$ and its projector P_t, write

$$\begin{aligned} h_\ell(t \mid x) &= P_t h_\ell(t \mid x) + z_\ell(t,x), \\ z_\ell(t,x) &= (I - P_t) h_\ell(t \mid x) \in \mathcal{R}_t^\perp. \end{aligned} \tag{36}$$

The decomposition is automatic; the hypothesis is not. It predicts that $P_t h_\ell(t \mid x)$ tracks token identity or engagement, while $z_\ell(t,x)$ carries most sense-disambiguating information. Sense labels should be more separable in $\mathcal{R}_t^\perp$ than in the orthogonal complements of matched control subspaces. For each sense C, the projected states $z_\ell(t,x)$ should also concentrate near the low-dimensional subspace $S_{t,C} \subset \mathcal{R}_t^\perp$.

Thus the critical question is not whether contextual embeddings move; they do. It is whether their sense-relevant movement has a privileged organization relative to a token-associated reference subspace, and whether conditional application of $\Pi_{t,C}$ explains that movement better than matched unstructured projectors.

IV. ATTENTION AND LATENT CONTEXT SELECTION

A standard attention head computes

$$h' = \sum_j \alpha_j \nu_j, \qquad \alpha_j = \frac{\exp(q_{\text{att}}^\mathsf{T} k_{\text{att},j}/\sqrt{d_{\text{head}}})}{\sum_{j'} \exp(q_{\text{att}}^\mathsf{T} k_{\text{att},j'}/\sqrt{d_{\text{head}}})}. \tag{37}$$

$q_{\text{att}}, k_{\text{att},j} \in \mathbb{R}^{d_{\text{head}}}$ are the query and key vectors of that head. The local head width d_{head} is an architectural quantity, not the usable carrier dimension d in the Goldilocks calculation. Keys and values are produced by different learned maps, so value vectors are not generally coefficients of a dual basis for the key vectors. Attention is therefore not literal basis reconstruction.

The useful analogy is softer. Attention routes information: it amplifies some contextual directions and suppresses others. In the terminology of Sec. III, this resembles context-dependent subspace selection. A literal basis-reconstruction interpretation would require additional constraints, such as approximately Parseval keys and values implementing the corresponding dual reconstruction. We do not assume this.

A compact latent-variable version is

$$p(y \mid x,t) = \sum_C p(y \mid x,t,C)\, p(C \mid x,t), \tag{38}$$

where t is the token being disambiguated, y is a possible output token, and C indexes candidate sense contexts. One may model

$$p(C \mid x,t) = \frac{\exp(s_{t,C}(x)/T)}{\sum_{C'} \exp(s_{t,C'}(x)/T)}, \tag{39}$$

with $s_{t,C}$ the gated score of Eq. (32). This is not a claim about the literal implementation of decoding in current LLMs: standard temperature acts on token logits, not on explicit latent sense labels. The point is only that ambiguity can be represented as a distribution over possible sense contexts rather than as a single fixed interpretation.

The same weights provide a soft implementation of the intertwined-observable construction:

$$\bar{h}_t(x) = \sum_C p(C \mid x,t)\, h_{t,C}(x) = \sum_C p(C \mid x,t)\, \Pi_{t,C} h(x). \tag{40}$$

Hard selection uses C^* from Eq. (33); soft selection uses Eq. (40). Attention may supply the routing signal $p(C \mid x,t)$, but the structured projectors or observables are an additional model component and should not be identified with standard attention weights without empirical evidence.

V. DISCUSSION

The geometric proposal is the capacity/accessibility distinction sharpened into the Goldilocks estimate (16). The intertwined-observable model is one possible contextual organization within that bracket, not a consequence of concentration geometry.

A. Relation to superposition and sparse autoencoders

The superposition hypothesis holds that neural networks represent more features than they have neurons by packing features into nearly orthogonal directions [5, 21]. The concentration estimate (5) gives the kinetic side of that picture: many directions can coexist. Equation (12) gives the epistemic side: many jointly active features create interference of scale $\sqrt{k/d}$. Sparsity is therefore not an implementation detail; it is necessary for readout.

Sparse autoencoders (SAEs) give a concrete empirical interface. They approximate layer activations as sparse expansions over learned decoder directions, making the w_i of Eq. (8) measurable objects rather than hypothetical features. Recent work has scaled SAEs, studied SAE feature geometry, and automated interpretation of large numbers of latents [22–25]. In the

TABLE I. Dynamic contextual embeddings versus the intertwined-observable hypothesis.

Dynamic description	Intertwined-observable hypothesis
Token state changes by context and layer	State is projected relative to $\mathcal{R}_t$
Layer activation carries context	$\mathcal{R}_t^\perp$ contains sense variation
Polysemy by vector deformation	Polysemy by conditional projection $\Pi_{t,C}$
Resource is depth and width	Resource is the Goldilocks bracket
Limit is activation bottleneck	Limit is crowding plus readout interference

present framework, SAE decoder directions can test the capacity/accessibility prediction, while sense-conditioned SAE activation clusters can test the stronger reference-subspace hypothesis.

B. A coactivation-weighted capacity limit

The active-set size k is a crude summary. A more refined description uses the coactivation graph

$$A_{ij} = \mathbb{P}(i, j \text{ active}). \tag{41}$$

If active amplitudes have, conditional on coactivation, zero mean and pairwise uncorrelated signs, then cross terms cancel in expectation and feature j contributes to the readout variance of feature i in proportion to $\mathbb{E}[x_j^2 \mid i, j \text{ active}](w_i^\top w_j)^2$. Absorbing amplitude variance and feature importance into coefficients b_i, b_j gives

$$E_{\text{int}} = \sum_{i \neq j} A_{ij} b_i b_j (w_i^\top w_j)^2. \tag{42}$$

Frequently coactive features are therefore pressured toward orthogonality. Rarely coactive or mutually exclusive features are much less constrained and may share directions or even become approximately antipodal if nonlinear readout makes this useful. The relevant object is not just the number of features, but the weighted coactivation graph.

C. Limitations

The framework can fail in several ways.

First, the Goldilocks bracket is a random-code estimate, not a model-independent hard boundary. Equation (15) assumes independent isotropic directions. Learned dictionaries can beat or underperform that null model. The bracket is informative only after the usable exact dimension d, dictionary count N_q, and tolerated overlap ε have been specified. The active count k belongs to the separate readout test.

Second, the nominal layer width need not equal the usable exact dimension d. Contextual embeddings can be anisotropic, and the occupied carrier can vary strongly by layer and training regime [26–29]. Whitening and related postprocessing can partially restore isotropic geometry [27, 28, 30]. At the same time, anisotropy should not be treated as an automatic or universal property of all Transformer representations [31]. The rule used to select or whiten a d-dimensional usable carrier must therefore be reported. We do not introduce a separate d_{eff}.

Third, the spanning matrix $U_{t,C}$ needs to be well-conditioned. If ρ in Eq. (29) is close to one, its dual amplifies readout noise.

Fourth, the reference-subspace pattern may simply be descriptively wrong. Trained models may represent polysemy without any preferred token-associated direction or subspace. In that case the capacity/accessibility account may still hold, but the reference-subspace hypothesis fails.

Fifth, the law $\sigma_{\text{int}} \sim \sqrt{k/d}$ assumes random-like incoherent directions and weakly correlated signs or amplitudes. Learned networks may improve on this by orthogonalizing frequently coactive features, or worsen it through anisotropy and correlated errors. The law is a scaling principle, not a universal equality.

Finally, the operator model has choices: the candidate reference subspace $\mathcal{R}_t$, the gate, the selector, the sense-subspace dimension, and the spectra in Eq. (25). The eigenvalues are not identified by the subspaces alone. These choices must be fixed or learned under held-out evaluation; otherwise the framework risks describing whatever geometry is found after the fact.

D. Empirical predictions

The key empirical task is to separate the geometric bracket from the stronger reference-subspace and intertwined-observable claims.

Claim G: a calibrated Goldilocks regime. For a prespecified overlap tolerance ε, a useful overcomplete representation should have $N_\mathrm{q} > d$ while its measured maximum overlap remains compatible with Eq. (15). Its accessibility should be tested separately against the active-set scaling in Eq. (12).

Claim A: privileged token reference subspace. For a polysemous token t, there exists a privileged token-associated subspace $\mathcal{R}_t$ such that sense-relevant variation is better exposed in $\mathcal{R}_t^\perp$ than in the orthogonal complements of matched control subspaces of the same dimension.

Claim B: low-dimensional sense subspaces. Within $\mathcal{R}_t^\perp$, sense-conditioned states concentrate near low-dimensional subspaces with small cross-sense coherence.

Claim C: intertwined context operators. Context projectors $\Pi_{t,C}$ share P_t, differ on $\mathcal{R}_t^\perp$, and improve held-out contextual prediction or compression relative to dimension-matched unstructured projectors. Where contexts are operationally incompatible, their fitted commutator should be nonzero and sequential projection should be order-sensitive.

These claims can fail independently. Claim G concerns the global dictionary and active-set geometry. Claim A can hold while sense information remains high-dimensional. Claim B can hold relative to some non-token-associated subspace, in which case Claim A fails. Claims A and B can hold even if the operator constraint in Claim C adds no predictive value.

The following tests separate the possibilities.

1. *Goldilocks calibration test.* Choose the usable carrier by a declared spectral threshold or whitening rule and report its exact basis size d. Count a prespecified dictionary of N_{q} feature directions, measure its maximum overlap, and estimate active-set size k. Compare the observed crowding and recovery behavior with Eqs. (15) and (12). The comparison must include isotropic random dictionaries and trained, coactivation-matched controls.

2. *Reference-subspace test.* For a polysemous token, choose candidate reference subspaces generated by the normalized static input embedding, mean contextual embedding, learned token-specific directions, or a low-dimensional combination of these. Project layer states as in Eq. (36). Sense labels should be better separated in $\mathcal{R}_t^\perp$ than in orthogonal complements of random, frequency-matched, or layer-matched control subspaces of the same dimension. The statistic may be classification accuracy or mutual information with sense labels, assessed by permutation over sense labels and bootstrap over token instances.

3. *Operational subspace test.* For each sense s, collect projected states $z_\ell(t,x) \in \mathcal{R}_t^\perp$ and estimate a low-rank subspace by PCA or probabilistic factor analysis. With $U_{t,s}$ the first q principal directions for sense s, measure cross-sense coherence $\|U_{t,s}^\top U_{t,s'}\|_{\mathrm{op}}$. True sense labels should yield lower cross-sense coherence than shuffled labels and stronger separation than matched control reference subspaces.

4. *Gated-readout test.* Linear probes trained on $z_\ell(t,x)$ should perform comparably to probes trained on the full state $h_\ell(t \mid x)$, while probes restricted to the reference component $P_t h_\ell(t \mid x)$ should perform near chance on sense classification.

5. *Intertwined-observable test.* Fit P_t and $B_{t,C}$ on training contexts, construct $\Pi_{t,C}$ and $A_{t,C}$, and evaluate Eq. (26) on held-out instances. Compare against unconstrained low-rank projectors, mixture-of- experts probes with the same parameter count, and shuffled context labels. Report $\|[\Pi_{t,C}, \Pi_{t,C'}]\|_{\mathrm{op}}$ and test whether reversing sequential context projections changes predictions where the model claims contextual incompatibility.

6. *Overlap-distribution test.* If a representation exploits quasi-dimensionality, pairwise inner products among token embeddings or SAE feature directions should concentrate near zero with variance $\sim 1/d$ after the usable carrier has been selected. Deviations quantify anisotropy or learned alignment not captured by the random null model.

7. *Synthetic concurrency degradation.* At a fixed layer, inject controlled superpositions
$$h_k = h_0 + \lambda \sum_{j=1}^{k} x_j w_j, \tag{43}$$
using directions w_j obtained independently, for example from SAE decoder directions or frozen linear probes. As k increases, readout should degrade smoothly on the scale $\sqrt{k/d}$ for random-like directions, with weaker degradation for feature sets that training has orthogonalized.

8. *Coactivation-dependent geometry.* Frequently coactive features should show reduced mutual coherence relative to rarely coactive features. This tests the broader capacity/accessibility framework through Eq. (42), independently of the reference-subspace hypothesis.

E. Scope of the quantum analogy

The analogy to quantum contextuality is structural and limited. Both settings can describe alternatives by observables with context-dependent eigenspaces. In the proposed construction, the operators $A_{t,C}$ intertwine because they share the spectral projector P_t; their complementary sense sectors may yield nonzero commutators. The hyperedges in Fig. 2 record only this incidence pattern, whereas Eqs. (23)–(25) supply its operator realization. A basis is a set of vectors, a subspace is their span, and an observable is an operator with a spectral decomposition; these objects are not interchangeable.

Quantum contextuality has the force of no-go theorems such as Kochen–Specker [32, 33]; no analogous theorem is claimed here for semantic embeddings. Real symmetric matrices can be noncommuting while still being stored and evaluated on an ordinary classical computer. Thus the operator construction is quantum-inspired, but noncommutativity by itself supplies neither Born probabilities nor physical quantumness.

Three levels of analysis should not be conflated.

1. *Classical high-dimensional representation.* A present-day Transformer carries real-valued hidden states through attention, nonlinear feed-forward maps, residual connections, and normalization. Concentration, quasi-orthogonality, feature superposition, dynamic embeddings, and finite readout are all available at this level. None requires quantum probability or quantum hardware.

2. *Quantum-like semantic or contextual formalism.* States, contexts, bases, interference terms, noncommuting questions, or measurement-like updates may be used as an abstract model of meaning even when the implementation remains classical. The relevant issue is not the vocabulary but whether this extra structure improves compression, explanation, or prediction relative to an ordinary contextual embedding model.

3. *Genuine quantum machine learning or quantum attention.* Here semantic states would be encoded in qubits or other physical quantum degrees of freedom, intermediate evolution would be unitary, phase would affect interference, and measurement would follow quantum probability. A standard Transformer does none of these things.

The Goldilocks bracket belongs entirely to the first level: it is a claim about classical high-dimensional geometry. The shared-subspace hypothesis is likewise testable in a classical model. The intertwined operators and their order-sensitive projections belong to the second level because they make the quantum-inspired comparison explicit and implementable. Their scientific value depends on the discriminative test in Sec. V D: they must improve prediction, compression, or explanation relative to equally expressive classical baselines. Even a positive result would support a useful quantum-like formalism, not physical quantum computation; phase-sensitive dynamics and Born-rule measurement are still absent.

The dimension of $\mathcal{R}_t$ is also explicit. In Hilbert-space quantum logic, different contexts can share one observable, as in the familiar tripod-like case, or several observables; a four-dimensional example has two contexts with two common observables [13]. The semantic analogue is $\dim(\mathcal{R}_t) = p$: $p = 1$ is the simplest case, while $p > 1$ permits several token-associated reference directions. In every case, sense-specific information is hypothesized to occupy subspaces of $\mathcal{R}_t^{\perp}$.

VI. CONCLUSION

Useful quasi-dimensionality is governed by two global counts. The exact dimension d is the size of an orthonormal basis of the usable carrier. The quasi-dimensional count N_{q} is the number of approximately orthogonal semantic directions represented in it. Only d is literally a vector-space dimension; N_{q} is the size of an overcomplete semantic dictionary. Their plain-language Goldilocks relation is

$$\boxed{d < N_{\mathrm{q}} \lesssim \exp\left(\frac{d\varepsilon^2}{4}\right)}, \tag{44}$$

where ε is the largest tolerable overlap. Below the left edge, an exactly orthogonal assignment remains possible. Above the right edge, random-like directions become too crowded for that overlap criterion.

This capacity statement is distinct from accessibility. If k directions are active together, their interference scales as $\sqrt{k/d}$. The symbol k is an active-set count, not a third dimension. Thus there is no universal "right dimension" independent of the dictionary, tolerated overlap, and readout task. For illustration, $d = 4096$ and $\varepsilon = 0.10$ give $4096 < N_{\mathrm{q}} \lesssim 2.8 \times 10^4$ under the isotropic random-code estimate; $k = 64$ gives an interference scale of 0.125. These numbers demonstrate the calculation but are not LLM constants.

The second central construction implements intertwined contexts. For a polysemous token, the context observables $A_{t,C}$ share the spectral projector P_t associated with token identity and differ on sense subspaces of $\mathcal{R}_t^{\perp}$. Their projectors $\Pi_{t,C}$ can fail to commute, and conditional projection $h_{t,C} = \Pi_{t,C} h$ supplies a context-specific embedding. A soft selector forms the mixture in Eq. (40). The "bank" schematic displays the financial, river-edge, and seat/bench contexts; the aircraft-banking sense is a possible fourth hyperedge rather than a replacement for the intended seat association.

These two ideas have different evidential status. The Goldilocks bracket is a classical geometric null model whose parameters can be measured directly. The shared reference subspace, the noncommuting context operators, and their predictive usefulness are stronger hypotheses. They must outperform dimension- and parameter-matched classical baselines on held-out data. A failure of the operator model would leave the capacity/interference bracket intact; a failure of the random-code bracket would reveal learned structure, anisotropy, or nonlinear readout not captured by the null model.

Finally, the operator construction is quantum-inspired, not a claim that a Transformer is a quantum device. It uses real matrices on classical hardware and introduces no Born rule, unitary dynamics, or physical measurement. Its value lies in whether shared spectral projectors, incompatible context operators, and order-sensitive conditional embeddings provide a more compact or predictive account of polysemy. That question is discriminative and falsifiable; the vocabulary of quantum contextuality is useful only to the extent that the proposed tests answer it positively.

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